

How Website Owners Use Consent Management Platforms: An Interview Study

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Abstract

Website owners must bring their consent interfaces into compliance with EU privacy and data protection regulation. Previous research indicates consent interfaces are often designed to manipulate users, violating consent requirements under GDPR. However, little empirical work has investigated how website owners configure, implement, and maintain consent interfaces, or what causes their (non)compliance. We conducted 12 interviews with website owners in Denmark from websites where their compliance had improved over the course of three years. Our results show that while they cared about privacy and legal compliance, most knew or cared little about their consent interface. Rather, the configuration and maintenance of the interfaces happened in collaboration with a diverse network of actors, with third-party consent management platforms playing a central role. We discuss the implications of this distribution of responsibility for regulators aiming to improve compliance, for legal attribution of controllership, and for researchers untangling the incentive structures and power dynamics surrounding online privacy.

CCS Concepts

• **Security and privacy** → Usability in security and privacy; • **Human-centered computing** → Empirical studies in interaction design; • **Social and professional topics** → Governmental regulations.

Keywords

Cookie, Banner, Pop-up, Consent Management Platform, online tracking, GDPR, Website Owners, Interview Study, Qualitative Research

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1 Introduction

The majority of consent interfaces on the European web – interfaces which establish a lawful basis for accessing or storing (personal) data from their visitors – do not meet the requirements for legally valid consent set out in the General Data Protection Regulation (GDPR) [23]. European regulators continue to struggle with enforcing these rules despite vocal and widespread concerns about the infringement of individuals' fundamental rights to both data protection and privacy, as well as the threat to states' electoral integrity and national security as a result of this online tracking [5, 27].

Although there is a considerable body of work that studies the prevalence of manipulative consent interface designs on the web [3, 16, 23, 26] and the effect they have on the behaviour of website visitors [1, 13, 24], there are few empirical studies that engage with the experience of the website owners who implement those interfaces. This paper presents early results from an ongoing multi-year project that investigates how consent interfaces change over time in response to regulatory action. The exploratory research question structuring this inquiry is thus: How do website owners - in practice - configure, implement and maintain Consent Management Platforms?

We use data from our research to identify websites that have used a consent management platform (CMP) for multiple years – third-party services which provide consent interfaces (and additional features) that have been widely adopted across the web since the GDPR went into effect [14] – and have changed their consent interface to be more compliant (i.e., add a "reject all" button on their first layer). We contacted 398 websites from this subset and present findings from a qualitative interview study about their reasons for this change and overall experiences maintaining the interface, conducted between December 2024 and January 2025.

2 Background

2.1 Consent interfaces

Multiple disciplinary communities study consent interfaces. HCI research focuses on user behaviour and perceptions through usability assessments. These studies show, for example, that users rarely engage beyond the first layer of the interface [4, 24], or that requiring an extra click to reject consent considerably increases consent rates [1, 13, 15]. Computer science often engages in more computational methods such as large-scale measurements of the overall prevalence of interfaces or specific designs. These indicate that consent interface prevalence doubled year over year from 2018 to 2020 [14], that currently roughly 50% of websites contain a consent interface [17, 26], and that the minority of them have a single-click

reject button [3, 17, 18]. Legal research has focused on interpreting data protection and privacy legislation and case law and what that implies for the legality of consent interfaces' design and technical functioning. For example, on the lawfulness of so-called "pay-or-consent" interfaces [22], on legitimate interest as a legal basis [21], on compliance with the purpose specification principle [9].

2.2 Consent management platforms

Consent interfaces first emerged in 2009 in response to the EU's ePrivacy Directive, which required explicit consent for any reading or writing of data on a person's device [8]. However, it was not until the GDPR went into effect in 2018, that consent interfaces became an inescapable quality of the (EU) web, and that a lively market emerged around providing such compliance services [14]. These so-called consent management platforms (CMPs) are software solutions designed to manage the collection, storage, and sharing of user consent signals [28]. While CMPs all share the same core functionality – providing a consent interface for the website visitor and a back-end for editing its content – they can vary quite drastically based on the other features they offer and the price they charge. Although no widely used typology exists, CMPs can arguably be divided into four categories [23]: *Comprehensive CMPs* offer various services, such as analytics and website scanning across different devices, as well as integrations with various platforms (e.g., UserCentrics and OneTrust). *Plugin CMPs* have more basic features and integrate with popular platforms such as WordPress, sometimes with optional paid features (e.g., CookieYes, Complianz). *Integrated CMPs* are solutions that are offered by a (eCommerce) platform themselves (e.g., Shopper, Shoptet). *Open source CMPs* are free scripts with only basic functionality and customisability (e.g., tarteaucitron, CookieConsent).

2.3 Website owners' experiences

An aspect that has been studied less is how CMPs end up on the websites and the website owners who implement them. Toth et al. [31] investigated the back-end interfaces and installation process of various popular CMPs and found non-compliant default settings that might nudge website owners to configure their interface in a consent-maximizing way, but did not interview or observe any website owners going through the process. Stöver et al. [30] did investigate the website owner's experiences, although not with CMPs. They analysed a dataset of 1043 responses to a notification about a non-compliant configuration of Google Analytics. Using thematic analysis, the researchers developed three personas that captured specific circumstances, contexts, and motivations: the ignorant hobbyist, the busy self-employed, and the informed multi-stakeholder.

To the best of our knowledge, no empirical work exists on the website owners' experience with CMPs.

3 Methods

We present findings from our qualitative interview study comprising 12 semi-structured interviews with 13 individuals from Danish websites who recently updated their CMP and were in charge of, or closely involved in, their consent interface and data protection policy.

3.1 Recruiting and Participants

We recruited participants by looking at websites with a CMP in 2020 without a reject button on the first layer and which still had a CMP in 2023 but now did have a reject button. We identified these websites by comparing the results of two scrapes of the top 30.000 most popular .dk-websites (using rankings [20]), from November 2020 and April 2023. The web scraper was inspired by Nouwens et al. [24] and focused on six popular CMPs: QuantCast, OneTrust, TrustArc, CookieBot, Crownpeak, and CookieInformation. Consent interfaces and buttons were detected based on CSS selectors used by those CMPs. We manually verified the detections by comparing the automated result with a screenshot. After filtering the websites for the continued use of a CMP but increased compliance, we were left with 624 domains.

We sent emails to 398 websites. For each website, we identified either a generic email address (info@domain.com) or an email address listed in the privacy policy of the responsible data protection officer (dpo@domain.com). We sent an email explaining the goal of our study and invited them for a 30-minute interview. We determined that 12 interviews were appropriate by assessing the amount of novelty and richness gained from the interviews [32], aligning with Guest et al. [12], who demonstrate that a sample size of 12 interviews is often sufficient to capture the majority of thematic codes within a relatively homogeneous group.

The participants worked in diverse organizational settings, including non-profits (4 interviews), for-profit businesses (5 interviews), and public institutions (3 interviews). Their seniority varied, with some having extended experience in their roles, while others had more recent involvement. They came from a range of professional backgrounds, including IT specialists, communications professionals, and compliance officers. Despite these differences, all participants were identified as being responsible for or involved in managing their organization's consent interface and cookies.

3.2 Limitations and Biases

Our recruitment process introduces several biases. First, by focusing on websites with a CMP installed, we inherently target website owners who have already considered privacy or compliance and are somewhat familiar with the subject. Second, the self-selection effect amplifies this interest, suggesting that our sample comprises more reflective website owners than the general population. Furthermore, we received several emails from website owners who expressed interest in the study but admitted they "did not know anything about cookies". This likely means that our participants represent an above-average interested website owner who has put in more effort and care to understand and comply with the regulation. The reader should assume that a whole segment of the population not interviewed is less engaged with the topic. In other words, these findings arguably only reflect the "best case" scenarios.

3.3 Interviews

The first author conducted 12 semi-structured interviews [19] with 13 website owners (as one interview had two participants), focusing on their work with the CMPs and how it had been implemented. 11 of the 12 interviews were conducted remotely using teleconference

software. One interview was conducted in person at the request of the website owner.

The interviews were conducted in Danish, recorded and lasted between 24 and 59 minutes (average: 36 minutes). Our interview guide included questions about their current CMP setup, recent changes to the consent interface, the observed change of the "reject all" button, and considerations of cookies and tracking technologies. In preparation for each interview, the first author visited the websites to inspect the CMP, cookies that were placed, and embedded frameworks (e.g., Google Tag Manager) and prepared questions based on this (e.g. "Why do you have two types of website analytics?").

3.4 Analysis

To analyze the interviews, we first generated a draft transcript using an institutionally-provided open-source speech recognition model (OpenAI's Whisper, v.3 [25]). We used the 'large' version of the Whisper model for transcription and WhisperX to attribute speech segments to individual speakers. We manually verified and corrected the transcripts.

The first author imported the transcripts into qualitative data analysis software MaxQDA and conducted a complete reading and descriptive coding of all 12 interviews. In this process, the first author coded the interviews using descriptive codes (e.g., "talking about abandoning Google Analytics"). These descriptive ("primary cycle") codes, gradually developed into more general emergent themes. The first and second author then held meetings discussing these emergent themes, which were then connected to current literature. We constructed the argument presented in this paper based on the insights from this coding and reflexive conversation as well as a review of the current literature.

To ensure anonymity, we do not introduce demographic information here, and all names of interview participants in the analysis are pseudonyms, and - for the context - irrelevant information has been altered.

3.5 Positionality

The first author is a trained qualitative researcher with a PhD in how humans make sense of technologies. They are interested in data protection, but less familiar with ongoing debates around CMPs. The second author has been engaged in academic conversations about online tracking and CMPs for the past six years. They are interested in uncovering how data protection can be improved. We actively used these different vantage points in writing the paper to align website owners' perceptions with current literature and debate.

4 Findings

4.1 Valuing Data Integrity and Compliance

All interviewees demonstrated an interest in the way their websites collected data and the status of their compliance with privacy and data protection regulation, likely because our recruitment approach resulted in the self-selection of individuals with an above-average interest and motivation on the topic (they have deployed a CMP for at least three years and agreed to an interview about it). Particularly, those representing non-profits and government institutions had

little interest in processing users' data against their will. Even website owners who "had not worried about [data protection] for a long time" still showed concern about data practices elsewhere in their business.

Many participants shared the sentiment that caring for user data was important. Even Frank, the interviewee most sceptical of the GDPR and data rights, acknowledged the importance of complying with legal requirements, and his business relied on the effectiveness of his advertising efforts that were based on user data.

4.1.1 Unwitting updates. However, most respondents found it difficult to recall specifics when the conversation turned to consent interfaces and the fact that they updated the buttons in it. They admitted to not having actively thought about or worked with the interface in recent years. Despite their general interest in compliance and data integrity, consent interfaces were often updated unwittingly, with little engagement from the website owners. Many responded with surprise when they were told that the content of their consent interface had changed over the last three years, and it was difficult to get an exact history of how the pop-up had ended up on their site.

Jens: The cookie banner was put up- I think it's five years ago. Five to six years. Yes.

Interviewer: We've detected a change within the last three years...-

Jens: Three years sounds just as good as five – it all blends... (Interview 7)

Jens, the webmaster of an association with "a couple of thousand members", had little idea of how his consent interface looked and referred to the issue of the consent interface as being "low interest". He was not sure when it was updated last and had little idea of how it currently looked. Similarly, when asked how he would access his CMP account, Johan, the owner of a webshop selling small machinery, stated:

Well, I just tried 10 minutes ago with my email and password, and it's not recognizing me... Now it says it's CookieBot, a part of UserCentrics - 'consent management made easy', but they've made it hard, I'd say. I'll probably have to reset... (Interview 3)

The process of trying to sign in to prepare for the interview revealed that CookieBot had been acquired by another company and left him feeling barred out. Since the website update a "few years back" nothing major had been done about the consent interface, as far as he recalled.

For Tom, the webmaster of a small sports association, the interview was also an occasion to check up on his CMP. During the interview, he accessed the CMP through his WordPress installation, an interface he had only accessed "very, very rarely" since setting it up during the website launch:

Damn- so many steps [...] We'll be there soon [...] Almost there [...] It's really something, you see? It really is... But it's a nice- easy interface. (Interview 2)

Tom proceeded to use the CMP feature that scans for new cookies on the website which should be added the interface and was pleased to identify that there were no additional cookies. It did, however,

leave him puzzled to hear that we observed a Google Analytics cookie being placed before the user consented:

Interviewer: So this Google Analytics tag... It looks like it's being loaded before [consent].

Tom: It's strange that it doesn't catch it. It could be because we've stuffed it in manually. [...] There's somewhere you can put some code in. It could be that it was shoved in there, and it should have gone through [the CMP]. It's possible. That... I'll make sure to get to that. (Interview 2)

Marie, the communications officer of a government institution in charge of the website, was similarly made aware that there were two different consent interfaces on their website, with one provided by the CMP and one supplied by a recently installed analytics software:

Marie: It says... Something with privacy. And then I think – is privacy the same as cookies?

Interviewer: It's also in the cookie banner-

Marie: We should be able to... Remove it within [analysis software](Interview 12)

She then used the interview as an opportunity to ask us about the various kinds of cookies and how her CMP worked. This also led her to explore the consent interface and what cookies the CMP had detected:

I can see, yes... Under marketing, it says 14. That's pretty wild. (Interview 12)

The website still had an active installation of Google Analytics even though she had recently made significant efforts to transition into another analysis platform that allowed them to keep their data "more secure" and within European territory. This demonstrates a common willingness to move away from free products towards more privacy sensitive solutions that several other interviewees also shared, but also how challenging it was in practice.

4.1.2 Purposive engagement. While the "unwitting updates" held true for most of the respondents, some participants purposively and regularly engaged with their CMP and consent interfaces. Two compliance consultants situated their CMP within their continuous compliance efforts:

It's gotten an increasingly greater focus as part of our yearly compliance wheel. (Interview 9)

They would hold biannual meetings about their consent interface and greatly consider it. For instance, they had discussed how to make a mobile-specific banner and worked together with the communications department on how that should impact the amount of text shown on the banner:

But I think... The actual text- she didn't delete much. There were a few things [...] for it to fit. But as a whole, we didn't delete much. (Interview 9)

Mike, an external data protection officer¹ for an NGO (among others), could refer to discussions with the authorities regarding the change to a "reject all" button:

I am pretty sure that it was something we did when the change in practice came. For a long period, we

probably ran with an "accept all"-button and an "save"-button, where some [purposes] were pre-checked. [...] and if it can help you [...] I have some emails on that (Interview 4)

In other words, while most could not analytically engage with the subject and appeared unaware of the changes, some held it as a primary focus. To some extent, this difference followed a pattern where the smaller organizations often lacked dedicated specific attention and resources.

4.2 Delegation of consent interfaces

A recurring motif among the interviewees was that they made reference to many other actors in complex, interwoven ecosystems that they relied on for the design of their consent interface. These actors range from compliance departments, umbrella organizations, SEO companies, marketing partners, freelance tech workers, and informal peer networks, each playing a distinct role in shaping how consent interfaces are designed, implemented, and maintained. A full untangling of the interplays is beyond our scope here, but we believe the identification (See Table 1) can be instructive to future research.

5 Actors, Roles, and Impacts on Consent Interfaces and Cookies

As Lea, the communication manager of a nation-wide NGO, sympathetically said towards the end of our interview:

So... [interviewer], I can also investigate a little more - if there's anything specific you'd like to know... I can always call our collaborators. (Interview 9)

Although we had been referred to Lea as the in-house expert on the matter, Lea felt she only held a limited understanding of the cookies they used and their consent banner. She trusted their "website partner", to whom she had described the interface during the website development years prior.

Website partner is just one of several actors that our interviewees referred to in relation to consent interfaces and cookies in general. Another often mentioned central actor was the *CMP provider*. Tom stated that their compliance was "fixed with the easy solution", referring to the CMP to whom they "paid a monthly fee". The CMP's weekly or monthly scans had motivated him to stop focusing on the consent interface and instead divert attention towards other parts of their responsibilities. This delegation to the CMP and other actors meant that updating the banner required little work for the websites themselves. Many interviewees were unaware of how their consent interface currently looked and what cookies they had on their website. Peter, a communications consultant eager to get away from Google products, investigated the consent interface on his website after it was pointed out that there was a YouTube cookie present.

Without Peter noticing, someone within the municipality had added a YouTube video to the website, and the CMP had scanned the website and added YouTube cookies to the banner. While the consent interface appeared legally compliant, the CMP did not aid in keeping up with the envisioned institutional policies around data protection and sharing. Similarly, Johan noted that he had not paid

¹An employee that functions as an internal regulator to ensure data protection compliance, which is a legal requirement under the GDPR for certain organisations.

Actor	Role	Potential Impact on Consent Interface and Cookies
Consent Management Platform	Provides software for implementing consent interface	Provides standards and occasionally offers consultancy, regularly scan sites
Freelance tech workers	Handle complicated programming tasks	Paid hourly and therefore consulted less frequently on issues that require them, yet are not considered urgent, such as the consent interface
Ad-data consultants	In charge of advertisement campaigns and collaboration with Google	Consulted when changes are made to the consent interface (e.g., updating tags) Call for tracking
Advertisement companies	Run advertisement campaigns to drive traffic	Request specific types of tracking, including server-side tracking, to achieve campaign goals
Journalist	Create content to increase audience engagement	Advocate for more tracking to understand users and produce tailored content
Communication department	Develop and oversee communication strategies	Responsible for updating and maintaining the CMP
Website vendor	Develop and sell “website”; maintain ongoing involvement	Often responsible for installing the CMS and CMP; typically the primary point of contact for major interface changes
SEO Company	Helps with content marketing plans and evaluates content performance	Request tracking cookies to optimize content and improve search visibility
Content Management Systems (Shopify, Umbraco, WordPress)	Provide a platform for the websites	Has built-in solutions for integrating CMPs facilitating integration through compatibility
Compliance department	Ensures GDPR compliance and mitigates risks	Advocates for reducing the number of cookies and collaborates with CMP providers
DPO	Data Protection Officer; oversees compliance efforts	Monitors and assesses risks associated with cookies and tracking mechanisms
Dansk Erhverv	Business Association in Denmark	Collaborates on compliance standards and provides industry-specific guidance
Sports Association Federation	Represents collective of associations and supplies technical and legal support	Influences policy and standards related to cookies and user consent
Statens IT	Umbrella organization of Public IT Solutions in Denmark	Provides IT services and advises on cookies. Sometimes in charge of CMP integration
Informal peer networks	Network of municipality workers who regularly discuss web topics	Offer feedback, and serve as a sounding board for interface improvements

Table 1: Actors, Roles, and Potential Impacts on Consent Interfaces and Cookies

much attention to his banner. He speculated that it could be part of a deliberate plan from the CMPs to operate without garnering much attention:

It’s a black box. It could be that it’s part of their business models [...] then it becomes like a fitness subscription you signed up to eight years ago... And you’ve kind of forgotten you have it, yet continue to pay, you know?

Although our interview data provided no insider information into the business models of CMPs, we would argue that if Johan’s theory holds, they are succeeding. Most of our interview questions about

how website owners stayed in tune with the authorities’ efforts turned out redundant; they rarely did.

6 Discussion and future work

Our study investigates the perspective of website owners with a recently updated consent interface, an oft-overlooked perspective in discussions of data protection and online tracking, where the focus is primarily on the experience of the website visitor [11, 13, 24] or the design of CMPs themselves [14, 31]. The website owners show a general tendency towards interest and care for their users’ data and their own legal compliance. However, when it comes to actual

discussions of the consent interface and the technologies involved, most show limited interest and capability. While our interviewees were self-described as the responsible person behind the consent interface, the actual work beyond the initial deployment was generally delegated to a complicated ecosystem of other actors with a central role for the CMP. These findings give rise to two considerations: the issue of controllership and the impact CMPs have on website owners' practices and how they impact data protection. We touch on these issues and connect them to what we consider alleys of future work.

6.1 Who is actually in control?

Our results indicate that the knowledge and maintenance work related to the consent interface, the collection of (personal) data on the website, and the requirements for compliance is not exclusively in the hands of the website owners themselves but that they instead rely on a broad network of actors and, as a result, website owners offload control and responsibility. This reality could have implications for both the legal standing of the website owners and for the strategic approach of regulators trying to fix non-compliant interfaces. Firstly, the question of who is in control carries with it legal implications: under the GDPR, the "controller" is the entity that determines the means (the "how") and purposes (the "why") of data processing and this designation comes with a host of obligations and liabilities [7]. Generally, controllership for online data processing is assumed to lie with the website owners, evidenced by the fact that fines for non-compliant consent interfaces in Denmark have so far only been given to websites, not "website partners" or "CMPs". Previous work by Santos et al. [28] has already called into question whether CMPs might actually be controllers when they scan for cookies or have non-compliant default designs. Our study on the experiences of website owners can be seen as yet another glimpse into the "factual reality" [6] of controllership. However, it should be noted that just because website owners do not "feel" in control and delegate the actual work to others does not necessarily mean they are not still legally controllers.

Secondly, regardless of the legal designation of website owners, if website owners delegate responsibility, it raises the question of whether they are the most effective target for regulatory action or whether strategies should be found to pressure the other actors. In Gillespie [10]'s classical argument on the "politics of platforms", he argues that while the notion of "platform" conveys a message of a neutral intermediary, the platforms are not neutral, and evoking it has a function of mitigating responsibility. In this case, the consent management *platforms* have an interest in remaining neutral tools for website owners to comply while avoiding liability. This distribution has arguably resulted in what Advocate General Bobek feared when he reflected on the danger of stretching the concept of (joint)controllership in a case where a website integrated a Facebook like button, namely that "[m]aking everyone responsible means that no one will in fact be responsible" [2]. If regulators center website owners in their theory of change and target their guidance and enforcement to them but neglect to consider the incentive structures and power dynamics they are entangled in, it might not be effective. Our study points to a vacuum of knowledge about these "factual realities" of website owners. HCI researchers,

through their deep engagements with human-machine interactions, and CSCW researchers, in particular, seem well-positioned to fill this void. CSCW researchers have, since the field's inception, developed and deployed a conceptual language for human sensemaking around collaboration in highly digitized environments such as articulation work [29] that we contend can prove valuable for this debate going forward.

6.2 Engaging with the path of least resistance?

Our study shows how CMPs aid website owners in their attempt to comply with current regulations, but do so unbeknownst to the websites. At a face level, this appears to be positive: website owners can comply with regulations and data protection without worrying about the technical details, and without fear, they might accidentally track individuals who have not given their consent. This leads to less stress, but also enables website owners to freely onboard new tracking technologies that feed large "consent coalitions" [33] while letting the CMP do the cookie scanning.

Many of the website owners in our sample, were only interested in minimal statistical data collection, for which a very basic server-side script would have probably sufficed, a solution which would significantly reduce their legal responsibilities. Instead, they often defaulted to data maximising solutions because these were the most user-friendly, most popular, or integrated seamlessly or by default with the CMP solution. Rather than investing research and regulatory effort into bringing consent interfaces into compliance, perhaps HCI research should explore these pathways of least resistance and regulators should try to increase friction around these solutions such that those who provide privacy and data protection by default and by design become the natural flow for website owners.

7 Conclusion

This study shows how website owners of websites with recently updated consent interfaces generally are interested in their use of data and their compliance with the law. Yet, many show a limited capacity to engage. Although responsible for website owners, consent management is delegated to a complex ecosystem of actors, of which the CMP is a central one. The study opens discussions of who is actually in charge of the data management, as although the website owners legally are responsible as controllers under GDPR, they often refrain from controlling the interface and instead trust other actors such as their CMP or website developer.

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